Quiz 1

Subject: Maths2 Maths2 16 Jul 26 Total Marks: 50.00 Questions: 14 With Answers

QUESTIONS

Q1 MCQ *[4.00 marks]*

Let A be a 3×3 matrix such that $\det(A) = 2$. Suppose B is a matrix obtained from A by swapping the second and third rows, and then multiplying the first row by -3. Which of the following options is correct?

OPTIONS

A. $\det(B) = 3$.

B. $\det(B) = 6$ ✓ Correct

C. $\det(B) = -3$

D. $\det(B) = -6$

Q2 MSQ *[4.00 marks]*

Let $M = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix}$. Choose the correct options.

OPTIONS

A. $\det M^T M \neq 0$ and rank $(M) = 1$.

B. $\det M^T M = 0$ and rank $(M^T M) = 1$. ✓ Correct

C. The matrix M is in the reduced row echelon form (RREF). ✓ Correct

D. The reduced row echelon form (RREF) of M is $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$.

E. rank $(M) \neq$ rank $(M^T M)$.

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Q3 MSQ *[4.00 marks]*

Consider the vector space

The subset

is a linearly independent subset of V. Which of the following matrices A are vectors in V such that $S \cup \{A\}$ is a basis for V?

$V = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \middle| a, b, c \in \mathbb{R} \right\}$. $S = \left\{ \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

OPTIONS

A. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ✓ Correct

C. $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ ✓ Correct

D. $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

E. $\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

Q4 SA *[4.00 marks]*

If $(7,5,3)$ is a linear combination of the vectors $(1,1,1), (1,-1,3)$ and (x,y,z) as $2(1,1,1) + 3(1,-1,3) + 4(x,y,z) = (7,5,3)$, then find the value of $x + y + z$.

ANSWER

0 (Equal • Numeric)

Q5 SA *[4.00 marks]*

Let A be an invertible (4×4)-matrix such that $A^T = A^2$. Find $\det A$?

ANSWER

1 (Equal • Numeric)

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Q6 SA *[4.00 marks]*

Let

If S forms a subspace of $M_{2\times 2}(\mathbb{R})$, then find the dimension of S. Otherwise, enter the number -1 as the answer.

$$S := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2}(\mathbb{R}) \middle| \begin{bmatrix} a & b \\ c & d \end{bmatrix}^T = - \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right\}.$$

ANSWER

1 (Equal • Numeric)

Q7 SA *[4.00 marks]*

Define a (3×3) -matrix $A = [a_{ij}]$ as $a_{ij} := \text{maximum}(i, j)$. Find the rank of the matrix A.

ANSWER

3 (Equal • Numeric)

Q8 SA *[4.00 marks]*

Consider the vectors $(k, 8, -5), (0, k-6, 4)$ and $(0, 3, k-2)$ in $\mathbb{R}^3$, for some $k \in \mathbb{R}$. Find the number of values of k for which the given three vectors are linearly dependent in $\mathbb{R}^3$.

ANSWER

2 (Equal • Numeric)

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Q9 MSQ *[5.00 marks]*

CONTEXT / PARAGRAPH

$M_{3\times3}(\mathbb{R})$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices:

Use the given information to answer the subquestions.

$$M_1=\begin{bmatrix}1&0&0\\0&1&1\\0&0&1\end{bmatrix}, M_2=\begin{bmatrix}0&1&0\\1&0&1\\0&1&0\end{bmatrix}, M_3=\begin{bmatrix}1&-1&0\\-1&1&0\\0&-1&1\end{bmatrix}, M_4=\begin{bmatrix}2&-3&0\\-3&2&-1\\0&-3&2\end{bmatrix}$$

Which of the following options are correct?

OPTIONS

A. $M_1-M_2+M_3=0$, where 0 denotes the zero matrix of order 3.

B. $2M_3-M_2-M_4=0$, where 0 denotes the zero matrix of order 3. ✓ Correct

C. $2M_2-M_1-M_4=0$, where 0 denotes the zero matrix of order 3.

D. The pair (a,b) is unique for which $aM_1+bM_2=0$ holds, where 0 denotes the zero matrix of order 3. ✓ Correct

E. There is more than one pair (a,b) such that $aM_1+bM_3=M_2$.

Q10 MCQ *[4.00 marks]*

CONTEXT / PARAGRAPH

$M_{3\times3}(\mathbb{R})$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices:

Use the given information to answer the subquestions.

$$M_1=\begin{bmatrix}1&0&0\\0&1&1\\0&0&1\end{bmatrix}, M_2=\begin{bmatrix}0&1&0\\1&0&1\\0&1&0\end{bmatrix}, M_3=\begin{bmatrix}1&-1&0\\-1&1&0\\0&-1&1\end{bmatrix}, M_4=\begin{bmatrix}2&-3&0\\-3&2&-1\\0&-3&2\end{bmatrix}$$

Which of the following options is true?

OPTIONS

A. $\{M_1,M_2\}$ is a linearly independent set. ✓ Correct

B. $\{M_1,M_2,M_3\}$ is a linearly independent set.

C. $\{M_3,M_4\}$ is a linearly dependent set.

D. $\{M_2,M_4\}$ is a linearly dependent set.

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Q11 MSQ *[5.00 marks]*

CONTEXT / PARAGRAPH

Let A be a coefficient matrix, and b be a column matrix to a system of m linear equations with n variables, $x_1, x_2, \cdots, x_n$, given by $Ax = b$; where $x = (x_1, x_2, \cdots x_n)^T$. Use this information to answer the given subquestions.

Let $C(A)$ denotes the column space of A. Choose the correct options.

OPTIONS

A. If $b \in C(A)$, then there exists vector $x \in \mathbb{R}^n$ satisfying $Ax = b$. ✓ **Correct**

B. If $b \in C(A)$, then there exists a unique vector $x \in \mathbb{R}^n$ satisfying $Ax = b$.

C. If there exists a vector $x \in \mathbb{R}^n$ satisfying the $Ax = b$, then $b \in C(A)$. ✓ **Correct**

D. If $b \in C(A)$, then the system of linear equations $Ax = b$ is consistent. ✓ **Correct**

E. If $b \in C(A)$, then the system of linear equations $Ax = b$ is inconsistent.

Q12 SA *[4.00 marks]*

CONTEXT / PARAGRAPH

Let A be a coefficient matrix, and b be a column matrix to a system of m linear equations with n variables, $x_1, x_2, \cdots, x_n$, given by $Ax = b$; where $x = (x_1, x_2, \cdots x_n)^T$. Use this information to answer the given subquestions.

Let

and

. Find the value of a, up to two decimal places, such that the system of linear equations $Ax = b$ does not have any solution.

$A = \begin{bmatrix} 3 & a \\ 4 & 1 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

ANSWER

0.74 – 0.76 (Range • Numeric)

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